Embedding

Every word in the vocabulary has a learned vector in a huge database. It is basically the raw meaning of a certain word.

- Each embedding is unique to each word.

The 3 weight matrices (W_Q, W_K, W_V)

Each word's embedding is multipled by the 3 matrices to obtain the 3 vectors: Query, key and Value.
The weight matrices are shared across all words for that specific layer.

The values of the matrices are learned by training:

1. At the start, every number in all the weight matrices is random.
2. The model reads some text and tries to guess the next word.
3. Checks if it is right
4. If wrong, every matrix gets tweaked slightly to get a better answer.
5. Repeated millions of times, until values produce good guess.

- Runs through all layers to get an output

Attention layers

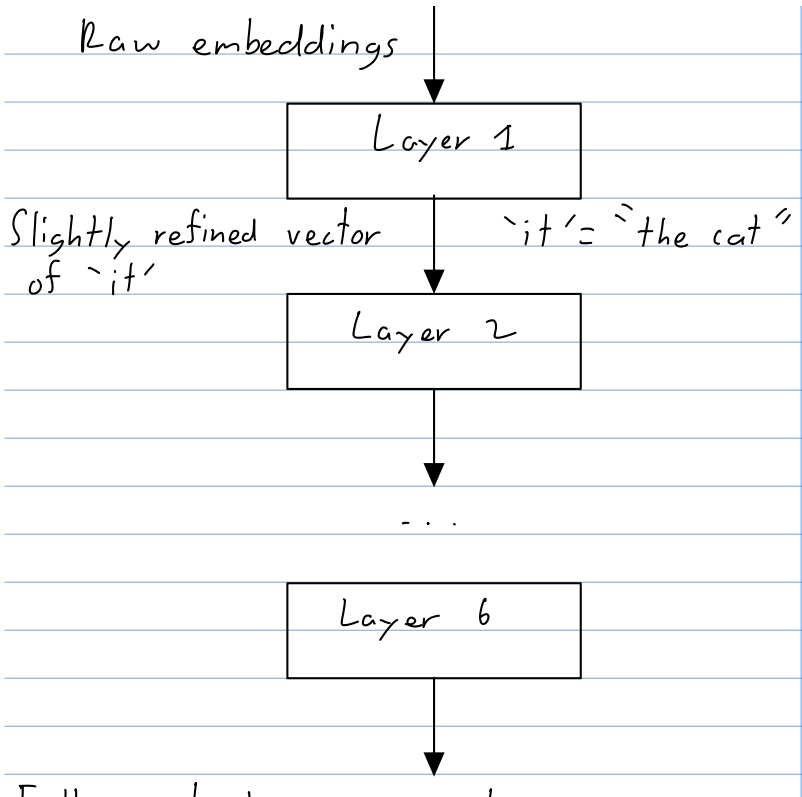
An attention layer = one full round:

- Every word forms their Query, Value, key.
- Every other word gets attended to
- A new context-enriched vector is outputed.

The transformer originally has 6 layers where each layer has their own unique weight matrices.

- Each pass refines the word
- The final output is a fully context-aware version which the model uses to predict the next word.

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The check

The model compares the single guess to the actual word that comes up after, to determine how wrong it is.

Backpropagation

The error is tracced backwards from layer 6 to 1. The model works out how much each matrix, to each individual number, contributed to the mistake. This is done by analysing how much would the final error change if you changed the number abit.

Then every number is slightly changed, changing each layer's weight matrices uniquely.